

Autonomous Space Anomaly Investigator (ASAI)

An Agentic AI Copilot & Automated Investigation System for Space Domain Awareness and Orbital Safety

The Core Concept: The same agentic architecture that investigates complex financial fraud, pointed at satellites instead of bank accounts. An autonomous AI investigator that detects suspicious orbital behavior, navigates relational knowledge graphs, actively tasks sensors to eliminate observational uncertainty, and recommends defensible next-best actions within strict safety policy boundaries.

1. The Operational Problem: The Fog of Orbit

Low Earth Orbit (LEO) and Geostationary Orbit (GEO) are experiencing an exponential population surge. With over 45,000 tracked objects, tens of thousands of mega-constellation satellites, and escalating debris fields, Space Domain Awareness (SDA) is overwhelmed. Flight Dynamics Officers (FDOs) and satellite operators face four acute operational bottlenecks:

- **Alert Fatigue from High-Covariance False Alarms:** Tracking radars produce coarse orbital state estimates with large error ellipsoids. Operators receive hundreds of Conjunction Data Messages (CDMs) weekly, over 95% of which are false alarms. Triage is manual, sluggish, and prone to human oversight.
- **Non-Cooperative & Covert Orbital Actions:** Adversarial or unannounced satellites perform stealth plane changes, rendezvous and proximity operations (RPO), shadow defense satellites, or eject uncatalogued secondary payloads without public registry notifications.
- **Fragmented and Siloed Space Intelligence:** Orbital ephemerides (TLE/OEM), radio frequency (RF) spectrum downlinks, optical photometric light curves, launch manifests, and operator registries exist in disconnected silos. Correlating them requires hours of manual cross-referencing.
- **Irreversible Decision Stakes (The Fuel Dilemma):** Every Collision Avoidance Maneuver (CAM) consumes irreplaceable onboard propellant, shortening the operational life of multi-million dollar assets. Operators cannot afford to maneuver on noise, nor can they afford inaction on genuine threats.

2. The Product Vision: The AI Orbital Safety Investigator

ASAI is an agentic AI system designed to act as a 24/7 intelligent copilot for satellite operations centers and space traffic coordinators. Rather than acting as a static dashboard or simple alert filter, ASAI is a **proactive, stateful investigative agent**. When an anomaly triggers a case, the agent explores the relational web of orbital assets, identifies what critical tracking data is missing, initiates controlled active sensing passes to collapse uncertainty, and formulates a defensible action plan with complete mathematical explainability.

3. Core Functional Pillars

- **Multi-Modal Anomaly Ingestion:** Detects triggers across four distinct channels: (1) Kinematic orbital residuals (unexpected delta-v, semi-major axis, or inclination shifts), (2) High-risk Conjunction Data Messages ($P_c > 1e-4$), (3) Signal anomalies (transponder silence, carrier drift, co-channel jamming), and (4) Operator investigation inquiries.
- **The Orbital Knowledge Graph:** Models space as a rich, interconnected graph: Satellites, Debris, Constellations, Operators, Launch Vehicles, Ground Stations, and RF Frequencies. Traversal algorithms detect coordinated swarm movements, co-orbital proximity, and parent-child deployment lineages.
- **Multi-Class Typology Assessment:** Classifies anomalies into operational event types: debris collision risk, non-cooperative RPO / shadowing, spacecraft bus tumbling (ADCS failure), intentional RF jamming, routine station-keeping drift, or novel/undocumented orbital maneuvers.
- **Active Information Gathering Under Uncertainty:** When initial tracking confidence is low, the agent does not guess. It formulates and executes a controlled, policy-compliant sensing request: tasking a robotic telescope for photometric light curves, requesting a precision radar pass to shrink covariance, or pinging an operator via coordination channels.
- **Two-Stage Decision Protocol:** Maintains dynamic belief states: (1) Baseline Recommendation generated prior to active sensing, and (2) Definitive Recommendation generated after new observations arrive, accompanied by a clear `what_changed` justification.
- **Human-in-the-Loop Policy Governance:** Enforces rigid operational guardrails. Autonomous execution is restricted to read-only queries and advisories. Critical propulsive burns (CAM) and formal ITU/UN dispute filings strictly require human flight director authorization.
- **Defensibility & Mathematical Stopping Rules:** Evaluates covariance volume reduction against Time-to-Closest-Approach (TCA) deadlines to know exactly when enough tracking evidence exists to act defensively.
- **Episodic Case Memory:** Persists resolved investigations into graph and vector memory. When similar maneuvers or anomalies emerge months later, the agent retrieves prior cases to accelerate triage.
- **Astrodynamic Audit Trail & Explainability:** Generates an auditable chain-of-custody for every recommendation: observations utilized, covariance reduction history, policy clauses cited, and mathematical justification.

4. The End-to-End Investigation Lifecycle

The system executes an explicit 7-stage cognitive loop designed to resolve ambiguity into defensible action:

Phase 1: Ingest & Open Case — Anomalous residual, CDM alert, or operator query triggers case creation with an immutable investigation UUID.

Phase 2: Graph Neighborhood Walk — Traverse relational graph to identify co-planar objects, operator ownership, launch manifests, and historical precedents.

Phase 3: Evidence Synthesis — Fuse ephemerides, radar cross-sections, photometric light curves, and space weather indices into an unified evidence graph.

Phase 4: Quantify Uncertainty — Calculate probability of collision (P_c via Foster algorithm), propagate covariance ellipsoids, and score assessment confidence.

Phase 5: Controlled Active Sensing — If covariance is too large to act defensively, issue a sensor tasking plan (optical pass, radar tracking, operator hail).

Phase 6: Recommend Next-Best Action — Re-evaluate post-observation covariance, evaluate flight safety policy rules, and generate the optimal action package.

Phase 7: Human Gate & Memory Commit — Present decision package to human flight director for sign-off, execute approved action, and anchor case in graph memory.

5. Technical Architecture & Component Stack

ASAI combines relational graph modeling, vector retrieval (GraphRAG), deterministic astrodynamics engines, and an agentic reasoning loop:

- **Agent Orchestration Engine:** Built on a modular multi-agent supervisor-worker architecture (LangGraph / State Machine). Maintains case state, manages iteration bounds, enforces stopping rules, and coordinates specialized tools.
- **Orbital Knowledge Graph:** Graph database (Neo4j / NetworkX / TigerGraph) storing space object entities and spatial-temporal relationships. Enables multi-hop queries: finding shared parent boosters, co-orbital clusters, and operator fleet patterns in milliseconds.
- **GraphRAG & Policy Vector Store:** Indexes unstructured spaceflight regulations, including NASA CARA ConOps, IADC Space Debris Mitigation Guidelines, ITU Radio Regulations, and operator Rules of Engagement (RoE). Grounding prevents LLM hallucinations.
- **Deterministic Astrodynamics Tools:** Deterministic computational tools exposed to the agent via Model Context Protocol (MCP) or standard tool interfaces: SGP4 orbit propagation, Foster 2D/3D collision probability (Pc), B-plane miss vector geometry, and minimum-delta-v impulsive burn planners.
- **Interactive Mission Control UI:** A comprehensive operator console: interactive 3D orbit visualization (Cesium / deck.gl), active case triage queue, orbital knowledge graph inspector, sensor tasking dashboard, and human CAM burn sign-off modal.

6. Real-World Public Data Pipeline

The platform ingests live and historical data from established, open public space domain repositories:

- **Space-Track.org / 18th SDS:** Two-Line Element (TLE) ephemerides, historical Conjunction Data Messages (CDMs), and SATCAT satellite catalog.
- **CelesTrak (Dr. T.S. Kelso):** Constellation ephemerides, space weather indices (Kp/Ap solar flux), and SOCRATES daily conjunction screenings.
- **Jonathan McDowell's GCAT:** Comprehensive General Catalog of Artificial Space Objects, detailing launch vehicles, parent-child stages, and bus models.
- **UCS Satellite Database:** Operational metadata: operator country, mission intent, dry mass, power generation, and design lifespan.
- **Open-Source Astrodynamics Libraries:** Skyfield, Orekit, SGP4, and Astropy for precise orbital propagation, frame transformations, and coordinate math.

7. Key Demonstration Scenarios (Showcasing the Tool)

To showcase the end-to-end power of the tool in a hackathon presentation or live demo, the platform features four rich, interactive operational scenarios:

Scenario 1: High-Covariance Conjunction -> False Alarm Resolved (Fuel Saved)

A commercial communications satellite receives a high-risk CDM showing a 120m miss distance, but cross-track covariance is massive (4.5 km). Rather than executing an expensive emergency thruster burn, the agent recognizes observational uncertainty, tasks an urgent radar re-observation pass, collapses the covariance, and proves the true miss distance is 4.2 km. The CAM burn is canceled, saving irreplaceable onboard fuel.

Scenario 2: Non-Cooperative Inspector Satellite -> Covert RPO Detection

An unannounced secondary satellite performs a discrete plane change into the orbital slot of a defense communications satellite. The agent walks the knowledge graph, correlates parent launch records, identifies a foreign state operator, detects coordinated low-thrust shadowing, and drafts an operator advisory along with a formal UN space object status report.

Scenario 3: Sudden Drag Anomaly -> Tumbling Spacecraft Triage

An Earth observation satellite experiences a sudden altitude decay and loss of telemetry. The agent tasks robotic ground optical sensors to collect a photometric light curve. Fast-Fourier transform analysis of the light curve reveals a 0.4 Hz brightness fluctuation, confirming attitude control (ADCS) failure and tumbling. The agent classifies the asset as an uncontrolled debris risk.

Scenario 4: Imminent Debris Conjunction -> Human-Approved Collision Avoidance Burn

A weather satellite is on a collision course with a defunct rocket body (miss distance < 50m, $P_c = 5.2e-3$, tight covariance). The agent computes the minimum-delta-v impulsive burn plan (0.35 m/s prograde), screens the post-maneuver trajectory against all cataloged objects to prevent secondary collisions, and routes the burn package to the human Flight Dynamics Officer for one-click authorization.

8. The Competitive Edge: Why This Wins

In a hackathon setting, standard AI projects present simple chat interfaces, static RAG search, or toy classifiers. **The Autonomous Space Anomaly Investigator wins because it delivers a complete, production-grade operational system:**

- **Solves a Real Trillion-Dollar Frontier Problem:** Addresses space traffic congestion, mega-constellations, and orbital collision risk with genuine domain rigor.
- **Principled Reasoning Under Uncertainty:** The system does not pretend to have perfect knowledge. It actively tasks sensors to collapse covariance before making high-stakes decisions.
- **Rigorous Physics + Graph + Agent Fusion:** Combines deterministic orbital mechanics (SGP4/Foster) with relational knowledge graphs and agentic decision intelligence.
- **Visually Stunning Mission Control Experience:** Features an interactive 3D orbit globe, real-time graph visualization, and human-in-the-loop flight control workflows.